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jmr&t Journal of Materials Research and Technology

journal homepage: www.elsevier.com/locate/jmrt

Original Article

Study on the effect of laser pre-sintering in laser-assisted glass frit bonding

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ARTICLE INFO

Article history:
Received 24 June 2022
Accepted 27 July 2022
Available online 4 August 2022

Keywords:
Laser technique
Glass frit bonding
Laser pre-sintering

ABSTRACT

With the rapid development of laser technology, laser-assisted glass frit bonding technology has been widely used in the packaging process of photoelectric devices. Despite its popularity, there still be some defects to affect the bonding performance, such as pores, cracks, non-fusion, and low joint strength. The glass frit should be pre-sintered by furnace or laser before laser-assisted bonding process. The behavior of the glass frit during the laser pre-sintering and laser bonding was studied in detail. It was found that the traditional pre-sintering process by a furnace can be replaced by laser pre-sintering when the thickness of glass frit layer was ca.3 μm. For the glass frit layers with thickness of ca.8 μm, a furnace should be used for pre-sintering firstly, and then the laser was used for re-sintering. The pre-sintering and re-sintering by laser can effectively inhibit the formation of pores in the following bonding process. The joined interfaces were inspected through a digital microscope and energy dispersive spectrometer (EDS). When the glass frit layer was re-sintered by laser, more penetration and less porosity was observed. It was found that the treatment of re-sintering by laser could improve the bonding strength obviously, the maximum bonding energy was 1.1 J/m^2. It can conclude that the laser plays a positive role in the pre-sintering process, and the bonding quality is improved fundamentally.

1. Introduction

Glass laser bonding is one of the common requirements for encapsulation technology of the electrical device, sealing of the vacuum glass and so on. Glass frit is generally formed by glass powder, solvents, binders, and fillers. Generally, the glass frit bonding steps are: (a) depositing a layer of glass frit on the glass substrates; (b) heat treatment of the glass frit on the glass substrates to remove the volatile additives, and thus consolidate and promote the adherence of the frit and the substrates; (c) bonding processing [1–3]. The bonding process would begin when the glass frit layer is heated to the bonding temperature. The glass frits with different compositions offer a wide range of bonding temperatures [4,5].

The most common method for glass frit bonding is thermocompression [6]. Generally, the bonding temperatures of glass

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https://doi.org/10.1016/j.jmrt.2022.07.162

Table 1 – Thermal properties of the material used in the study.

Material	Annealing point T_a/°C (10^{12} Pa s)	Transition temperature T_g/°C ($10^{11.3}$ Pa s)	Littleton softening point T_{lsp}/°C ($10^{6.6}$ Pa s)	Dilatometric softening point T_{dsp}/°C (10^8 Pa s)	Coefficient of Thermal Expansion α/ (10^{-7} °C^{-1})
Glass substrate	722	–	971	–	31.7
Glass frit	–	343	–	428	48

frits are higher than 380 °C, and the substrates should be heated to the same temperature inside the furnace. Therefore, the applications of the thermo-compression method are limited to encapsulation of photoelectric devices (e.g. organic light emitting diode displays) that can withstand temperature up to bonding conditions [7,8].

Except thermo-compression method, local melting of the glass frit by laser can also achieve the connection process. The advantages of laser-assisted bonding process are high energy density, small heat affected zone, non-contact, fast bonding speed and high efficiency, which can effectively solve the problems of high temperature furnace bonding [9,10]. Laser-assisted glass frit bonding has been investigated by more and more researchers. The detailed description of a laser-assisted sealing method can be found elsewhere [11–13].

In the process of laser-assisted bonding which uses glass frit as the intermediate layer, it is inevitable to pre-sinter the glass frit layer before bonding. The most common method of pre-sintering is to put the coated glass substrate into a heating furnace and heat the specimen to the pre-sintering temperature [14]. An appropriate increase in pre-sintering temperature can effectively level the glass surface and decrease the surface roughness, and thus the surface of glass melts becomes smoother [15]. The thermal process has been optimized to remove possible formations of bubbles/voids with three temperature steps, including drying, burn-out and pre-sintering [16]. The melting process of glass frit [17] and the total heat flux [18] will affect the formation of voids. Through the optimized thermal process, namely three-step bubble/voids removing pre-sintering process: 450 °C for 30 min in the regular air environment; 450 °C for 30 min under the vacuum environment and 500 °C for 60 min in the regular air environment, the processed glass frit layer is free of bubbles/voids [19]. In addition, quasi simultaneous laser bonding had been used after pre-sintering, because of a smooth simultaneous heating and cooling of the whole seam, the formation of voids and cracks was inhibited to some extent [20]. However, all of these commonly require a heating process in a furnace that cost generally more than 2 hours. To completely eradicate the pores in the connection process requires more complex heating methods and takes a longer time.

A method of laser pre-sintering was presented in this paper. Under the condition of low glass frit layer thickness, the organic solvent and adhesive were removed by a furnace. Instead of a pre-sintering step in a furnace, the laser was applied to finish the pre-sintering. It took less time than furnace pre-sintering, and the glass frit layer finished by laser pre-sintering had a better connection effect and lower porosity. As the thickness of the glass frit layer increases, the pre-sintering needs be completed in a furnace firstly, then the re-sintering by laser was used, which can reduce the porosity dramatically.

2. Material and methods

2.1. Materials

The glass substrates were alkaline earth boro-aluminosilicate glass (EAGLE XG, corning), the size was 40 mm × 40 mm × 0.7 mm. The glass frit was a low melting temperature paste (from BASS). The thermal properties of the material used in this study were shown in Table 1.

Malvern Master Sizer 2000 was used to measure the particle size of the glass frit. The average size was 1.42 μm, and

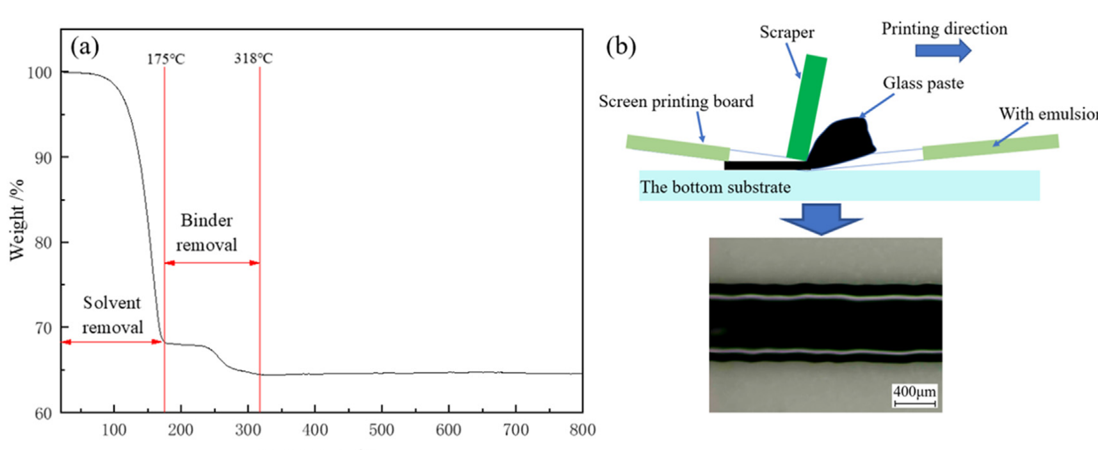

Fig. 1 – Thermogravimetric analysis (TGA) of the glass frit and the diagram of screen-printing process.

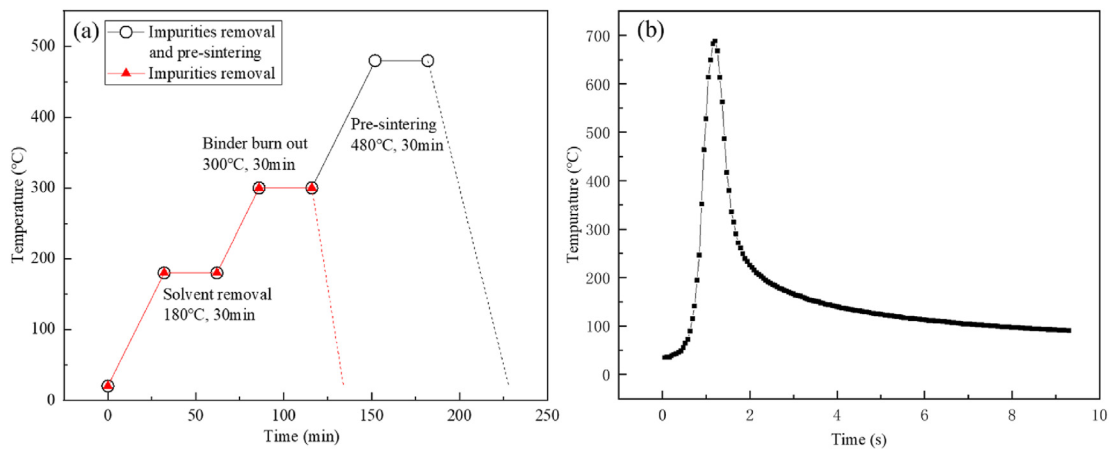

Fig. 2 – Heating process curve: (a) Heating process in a furnace; (b) Heating process by laser pre-sintering.

50% of particle sizes were less than 1.30 μm. The mass fraction of solids in the glass frit was 64.1%. The viscosity measured by Brookfield DV2 is 101 kCps (25 °C). The thermogravimetric analysis (TGA) of the glass paste was presented in Fig. 1(a). The TGA was carried out for a temperature range from 25 °C to 800 °C under air atmosphere, the heating rate was 5 °C/min. The TGA showed two major weight losses: the step between 0 and 175 °C was related to solvent removal; the step between 175 °C and 318 °C was related to binder removal.

2.2. *Sample preparation*

The glass substrates were ultrasonically cleaned with the mixture of distilled water and detergent, and the glass paste was screen-printed on the bottom substrates with screens, as shown in Fig. 1(b). Two types of screens were used, 400 mesh screen (the wire size: 18 μm, the thickness of emulsion: 10 μm) and 325 mesh screen (the wire size: 28 μm, the thickness of emulsion: 20 μm). The straight line (length: 30 mm; width: 0.8 mm) was printed on the glass substrates.

2.3. *Heating process*

The heating process was conducted first to remove unwanted components in the glass paste and then to pre-sinter it before bonding. The heating process usually includes three steps: solvent removal, binder burn out and pre-sintering [7], as shown in Fig. 2(a). Samples were briefly leveled at room temperature for 30 min before heating. The first two steps need be finished in a furnace, the heating rate was set as 5 °C/min. The pre-sintering process of the glass paste plays a critical role not only in the adhesion of the paste to the substrates but also in the quality of sealing after the bonding process. The pre-sintering step can be finished through three methods as followed.

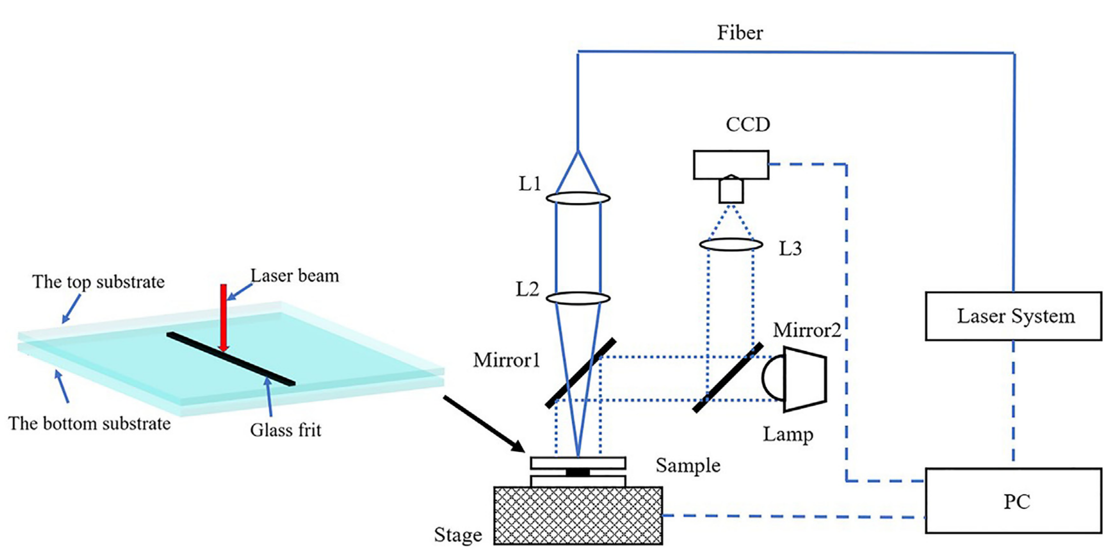

Fig. 3 – The diagram of laser-assisted bonding apparatus.

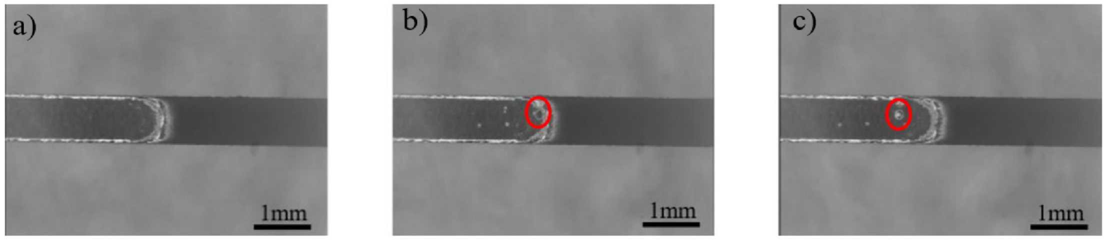

Fig. 4 – Laser pre-sintering process: (a) t = 10.07 s; (b) t = 10.10 s; (c) t = 10.20 s.

1. The printed samples were pre-sintered in the same furnace after the first two steps. The heating rate was set as 5 °C/min, and there was no control of the cooling down step. The burning out step removed all the additives in the paste. The pre-sintering step makes the paste compact and ready for bonding. The final thickness of the sealing glass paste after pre-sintering was ca.3 μm (400 mesh screen) and ca.8 μm (325 mesh screen).
2. The printed samples were pre-sintered by laser after the first two steps were finished in a furnace. The heating curve of the first two steps (i.e., solvent removal, binder burning out) was still the same as shown in Fig. 2(a). Then the samples were taken out of the furnace and the glass paste was pre-sintered by laser, the scanning speed and defocusing distance were 0.1 m/min and −15 mm. The temperature curve of the laser heating process was measured by an infrared thermal imaging camera (FLIR A615) as presented in Fig. 2(b). The maximum temperature during laser pre-sintering was 688 °C, which is much higher than that of furnace pre-sintering. The remaining pin-hole in the glass paste was got rid of more sufficiently.
3. All three steps (solvent removal, binder burning out and pre-sintering) were carried out in a furnace, and then the printed samples were taken out of the furnace and the glass frit layer was re-sintered by laser. The heating curve in the furnace is shown in Fig. 2(a) and the laser heating curve is also the same as shown in Fig. 2(b). The glass frit was pre-sintered sufficiently.

The surface morphology of pre-sintered specimens was observed by microscope (Primostar) and 3D optical profilers (Wyko NT9100), and the section contour was obtained. The micro pore morphology and the cross-section were observed using a scanning electron microscope (SEM, TESCAN MIRA3 LMU).

2.4. *Bonding procedure*

Laser test platform was built, as shown in Fig. 3. RFL-DDL-100 semiconductor laser was used, with a maximum output power of 100 W and the beam radiation wavelength is 915 nm. The focal length of the laser head is 150 mm and the spot diameter at the focal point is 300 μm. The laser bonding process was directly photographed by the coaxial CCD camera system with a frame rate of 545 frames per second, the resolution is 800 × 600. The surface morphology of the bonding layer was observed by a digital microscope (VHX-6000). The cross-sections of the specimens were observed by SEM. The chemical composition of the specimen section was determined by energy dispersive X-ray analysis (EDS, TESCAN MIRA3 LMU).

3. Results and discussion

3.1. *Laser pre-sintering and bonding processes photographed by CCD*

The pre-sintering process in a furnace is relatively mild, and there may still be voids in the glass frit layer. The laser pre-sintering process can further reduce the voids in the glass frit layer. When the laser radiated on the glass frit, the frit melted

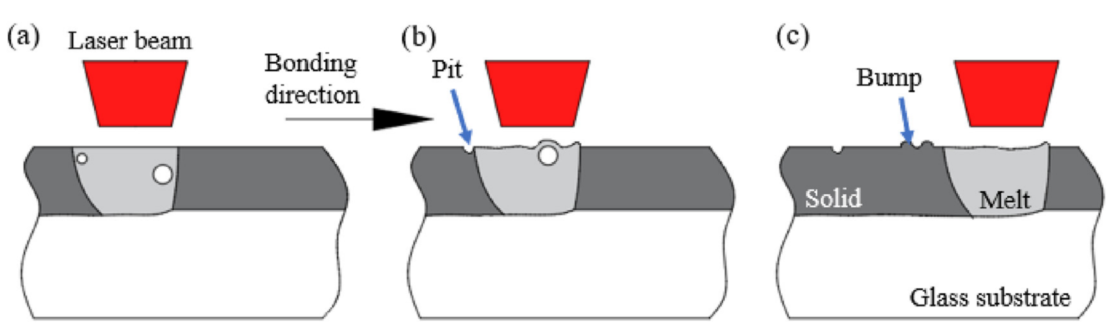

Fig. 5 – Schematic diagram of laser re-sintering process (cross-section): (a) Forming bubbles; (b) Forming pit; (c) Forming bump.

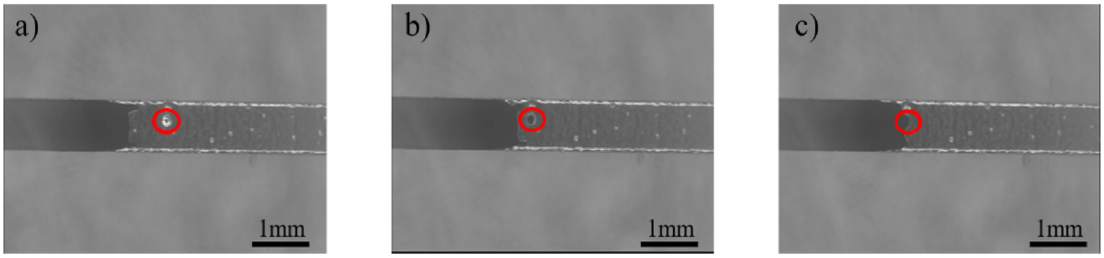

Fig. 6 – Laser bonding process: (a) t = 11.01s; (b) t = 11.09s; (c) t = 11.13s.

and the bubbles floated up. After the laser left, the glass frit solidified rapidly, and at the same time, the tiny bubbles which could not overflow completely would form pits on the surface of the glass frit as the glass paste solidifies. Larger bubbles that could not overflow completely would lead to appear of the bump. Fig. 4 shows the formation of the bump photographed by coaxial CCD, and its schematic is shown in Fig. 5. When the big bubble floated up, the bubble lifted the molten glass frit to form the bump. Then the bubble burst and the bump lost the bubble's support. With the solidification of the molten glass frit, the bump was formed finally. The appearance of the bump prevented the upper substrate from fully contacting the glass frit layer. In the actual connection process, the pits and bumps would not affect the bonding surface.

Fig. 6 shows the connecting process of the specimen after laser pre-sintering. The white points of the glass frit in Fig. 6(a) are small pits, and the larger defect marked with the red circle is a bump. The dark black area left was the bonding part of the glass frit and the upper glass substrate. As shown in Fig. 7(a), the substrate was bent downward and deformed. In Fig. 6(b), the small pits were directly covered by molten glass frit, and the bump melted before the molten glass frit front reaches it. Because the bump produced the color center and had a stronger laser absorption rate than the surrounding glass frit, as shown in Fig. 7(b). In Fig. 6(c), the molten bump was covered by the melting front. Then, as shown in Fig. 7(c), the clearance was filled with molten glass frit.

3.2. *Pre-sintering quality*

The samples with a low thickness glass frit layer (ca.3 μm) were studied. The sample as shown in Fig. 8(a) was pre-sintered inside a furnace, and the pre-sintering temperature reached 480 °C. The glass frit was not melted completely, and it had a rough surface. The cross-section contour in Fig. 9(a) shows that the thickness of the glass frit layer fluctuates in a range of 1.3 μm and the average height is 3.519 μm. Then, the sample was directly pre-sintered by laser. With the increase of laser pre-sintering power, different pre-sintering states of the glass frit appeared. When the absorbed laser energy was seriously insufficient, the glass frit could not be melted. When the absorbed laser energy increased continually but still not enough (P = 28 W), the pre-sintering was not sufficient, and its morphology and cross-section are shown in Figs. 8(b) and 9(b), respectively. The glass frit was still not melted fully, a smooth saddle surface was formed, and the dense pits mainly appeared at the bilateral thickest parts. When the absorbed laser energy was sufficient (P = 29 W), it was in a state of full pre-sintering, as shown in Fig. 8(c). The surface morphology is shown in Fig. 9(c). Because of the sufficient laser energy, the melted glass frit spread out on both sides, and the minimum height of the saddle surface (1.91 μm) was lower than that shown in Fig. 9(b) (2.53 μm), and the pores overflowed fully. After absorbing too much laser energy (P = 33 W), it was in an over sintering state, as shown in Fig. 8(d). The glass frit was in an unstable state, the particles inside the glass frit were shifted, some components were biased, and the phase split. The area where frit accumulated was higher than the surface, and other areas lack material. It was no longer suitable for bonding. If the laser pre-sintering power is near 33 W, the phase separation trend is stronger. To eliminate the phase separation and fully remove the pores and impurities in the frit, 29 W was selected for laser pre-sintering.

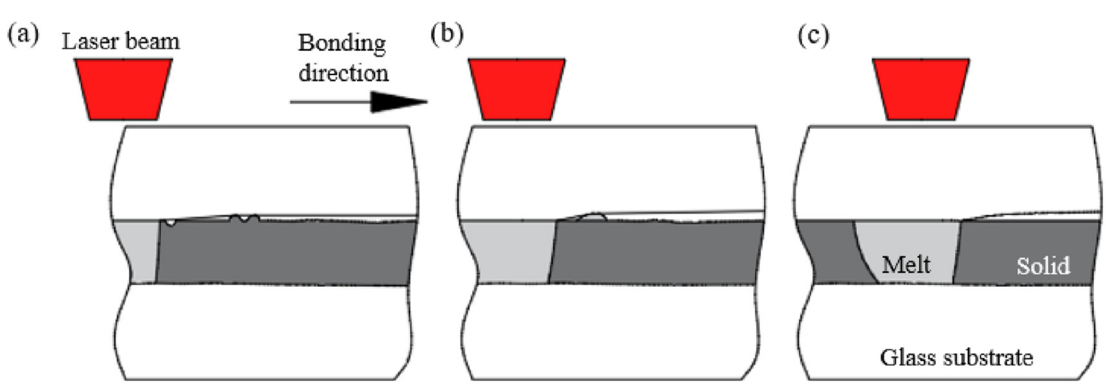

Fig. 7 – Schematic diagram of laser bonding process (cross-section): (a) Bonding; (b) Removing pit and remelting bump; (c) Gap filling.

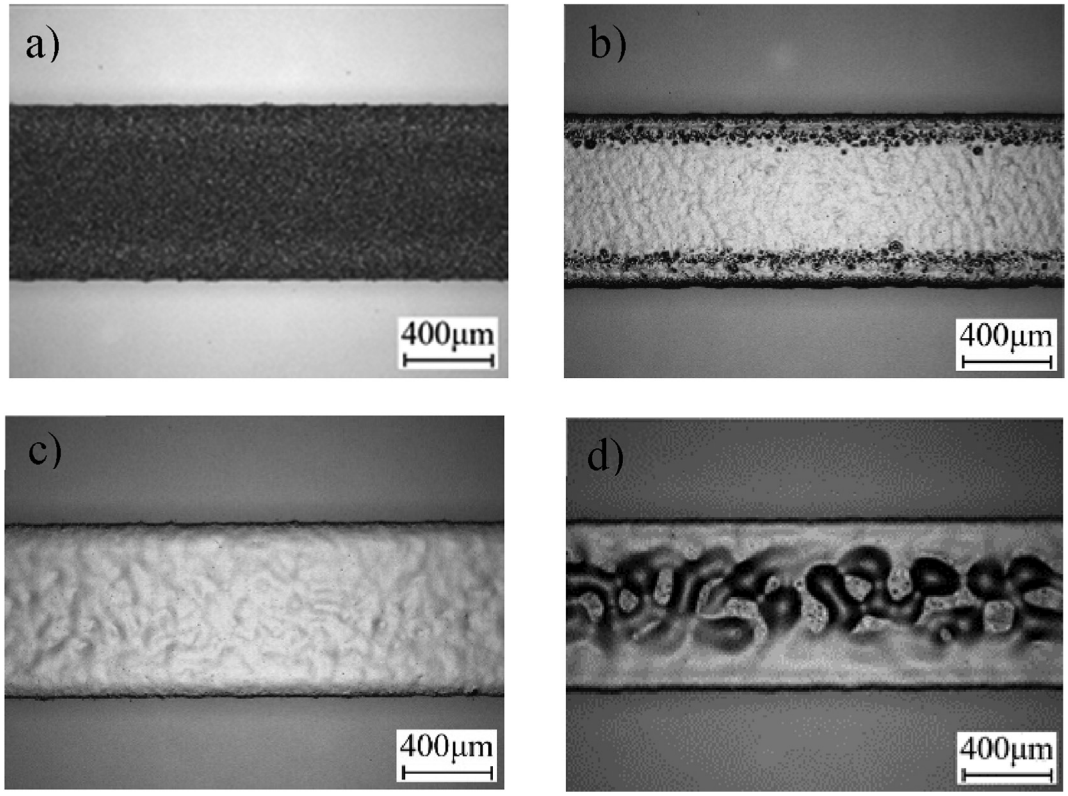

Fig. 8 – Surfaces of the different glass frit layers (ca.3 μm): (a) Pre-sintering in a furnace; (b) Insufficient pre-sintering by laser; (c) Full pre-sintering by laser (d) Over-sintering by laser.

Increasing the thickness of the glass frit layer (ca.8 μm), the situation became more complicated. The sample pre-sintered inside a furnace was still not melted fully and the surface was rough, as shown in Fig. 10(a). Its surface morphology is shown in Fig. 11(a). The slope of the glass frit edge was lower than that in Fig. 9(a). Then, the samples directly pre-sintered by laser were studied. Inevitably, with the increasing of the glass frit layer thickness, the glass frit produced more stress in the laser pre-sintering process. The large stress, lead to vertical grain in the glass frit, namely the crack appeared, as shown in Fig. 10(b) (P = 24 W) and Fig. 10(c) (P = 26 W). Based on the specimen pre-sintered in a furnace first. The re-sintered process by laser (P = 26 W) was studied, and the full re-sintering result is shown in Fig. 10(d). Compared with the results of direct pre-sintering by laser, the tensile stress of the glass frit layer was lower, and vertical stripes were not easily formed. If the laser energy was insufficient (P = 24 W), the saddle surface of the glass frit layer also existed during the re-sintering process, just as shown in Fig. 11(b). The surface of Fig. 10(d) with sufficient re-sintering (P = 26 W) was relatively flat, as shown in Fig. 11(c).

The surface of the pre-sintered glass frit was observed with SEM. The sample pre-sintered in a furnace had a rough surface and pin-holes, as shown in Fig. 12(a). Enlarged the magnification, the pin-hole was observed and extended deeply into the glass frit. In Fig. 12(b), the sample sufficiently

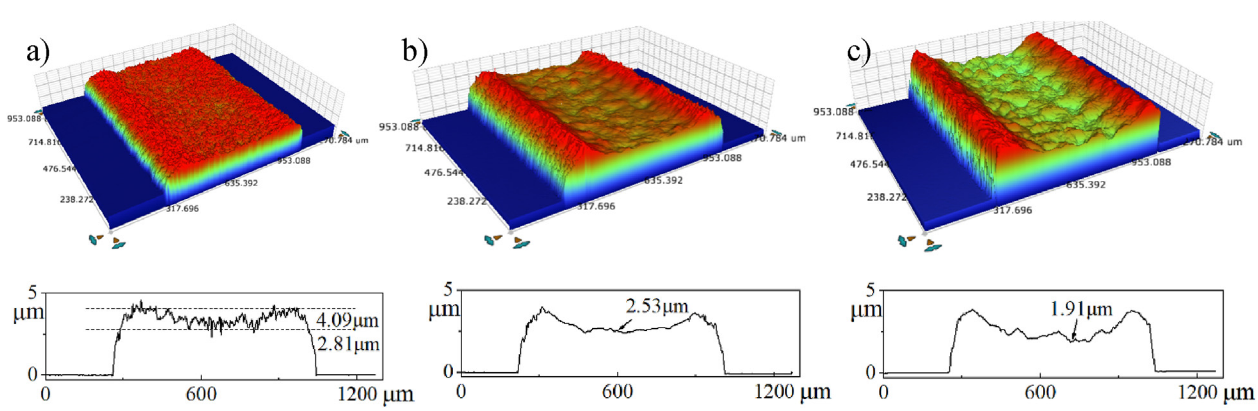

Fig. 9 – Surface morphology of the different glass frit layer by 3D optical profilers (ca.3 μm): (a) Pre-sintering in a furnace; (b) Insufficient pre-sintering by laser; (c) Full pre-sintering by laser.

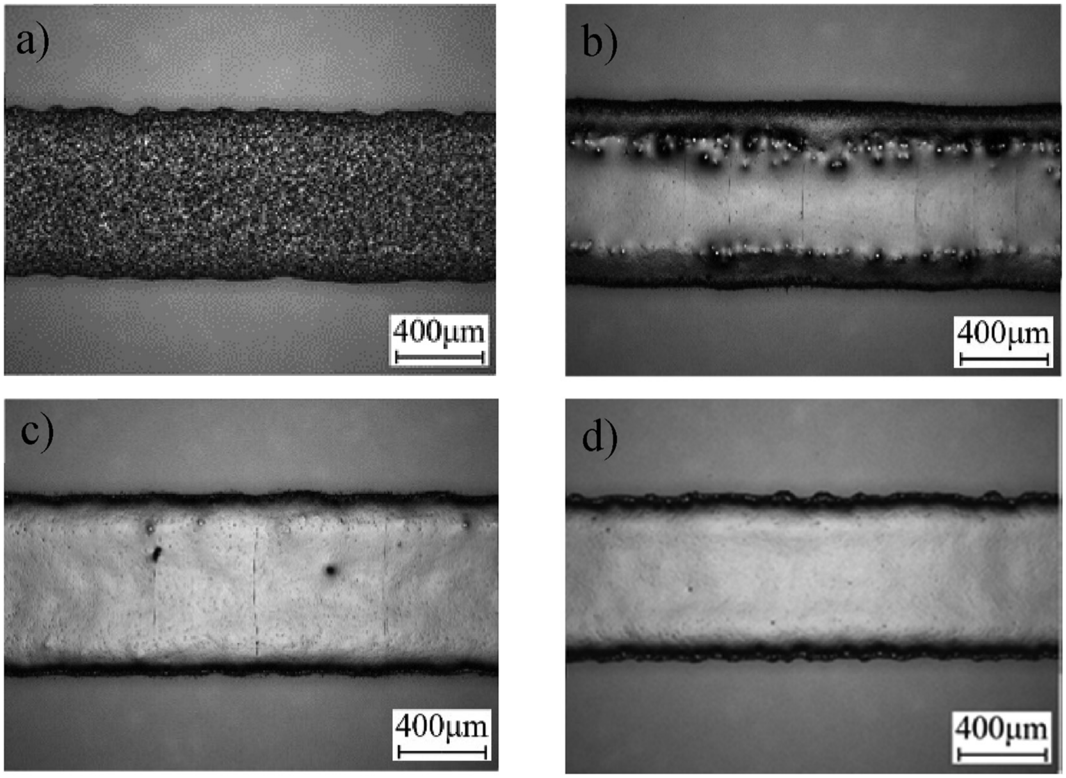

Fig. 10 – Surfaces of the different glass frit layers (ca.8 μm): (a) Pre-sintering in furnace; (b) Laser direct insufficient pre-sintering; (c) Laser direct sufficient pre-sintering; (d) Full re-sintering by laser.

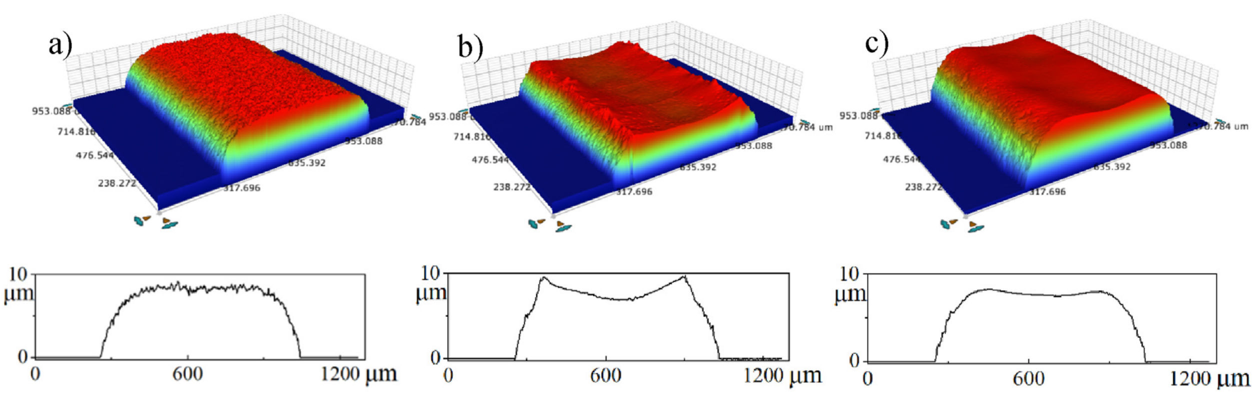

Fig. 11 – Surface morphology of the different glass frit by 3D optical profilers (ca.8 μm): (a) Pre-sintering in furnace; (b) Insufficient re-sintering by laser; (c) Full re-sintering by laser.

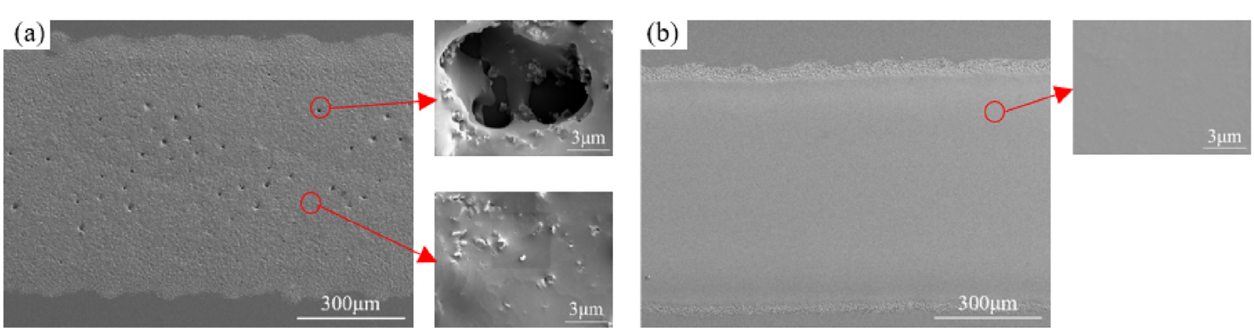

Fig. 12 – Surface morphology of the glass frit layer by SEM (ca.8 μm): (a) Pre-sintering in a furnace; (b) Full re-sintering by laser.

Table 2 – Table of test parameters.

Group	Printing thickness	Pre-sintering	Laser power P (W)	Bonding speed V (m/min)	Defocusing distance F (mm)
1	ca.3 μm	F	33、35、37、39、41、43	0.1	−15
2	ca.3 μm	L	40、42、44、46、48、50	0.1	−15
3	ca.8 μm	F	29、31、33、35、37、41	0.1	−15
4	ca.8 μm	F-L	32、34、36、38、40、42	0.1	−15

exhibits smooth surface and tight pre-sintering effects, indicating it was sufficiently re-sintered by laser. The only downside is that the crack appears easily if pre-sintering directly by laser without the pre-sintering process in a furnace, when the thickness of the glass frit layer is larger.

3.3. *Bonding quality*

After pre-sintering, the specimens were used for the bonding test, and the test parameters are shown in Table 2. The specimens pre-sintered by furnace were marked as “F”. The test pieces pre-sintered directly by laser were denoted as “L”. The specimens pre-sintered by heating furnace firstly, and then re-sintered by laser were denoted as “F-L”.

Pores in the connection area is a common defect of the laser bonding. The porosity content P (%) was quantified with the following formula:

$$P(\%) = S_1/S \tag{1}$$

where S_1 is the area of pores and S is the connection area.

The bonding test with the low thickness of the glass frit layer (ca.3 μm) was carried out under the parameter group 1, the results are shown in Fig. 13(a)–(c). When the bonding laser power was 33 W, only the middle part of the glass frit was connected, and the connection width was 652 μm. Micro pores appeared in the connection area, and the porosity was 3.82%. When the power increased to 37 W, the glass frit layer was completely connected. But a large area of pores appeared in the middle of the glass frit, and the porosity was 26.05%. When the laser power increased to 43 W, the width of the connection area was greatly increased, and the connection width was 1135 μm, more than the printed-screen width, because of the capillary force between the glass substrates. The pores were densely distributed in the bonding area, and the porosity was 59.98%.

The samples directly pre-sintered by laser (29 W) were used for the bonding test under the parameter group 2, and the results are shown in Fig. 13(d)–(f). When the bonding laser power was 40 W, only a narrow area on both sides was not connected, the reason may be that the glass frit after laser pre-sintering had a saddle shape with bulgy sides. The bonding width was 745 μm. The pores were not visible. When the power increased to 46 W, the glass frit layer was completely bonded and diffused exceeding the printed-screen width. Some small pores appeared sporadically in the bonding area, and the porosity was 0.21%. When the laser power was set to 50 W, the width of the connection area greatly increased to 1048 μm. Micro pores were mainly distributed in the central zone of the bonding area, and the porosity was 3.59%.

The effect on the bonding width of different laser power is shown in Fig. 14(a), The bonding width of the samples increased with the increasing laser power. The samples pre-

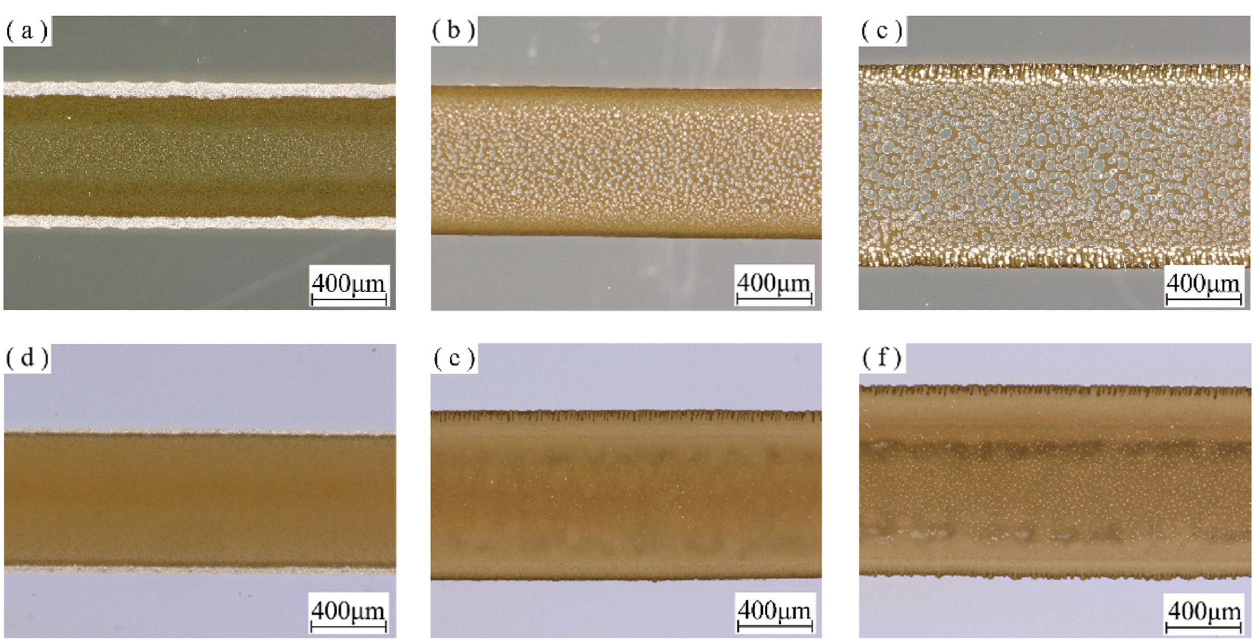

Fig. 13 – The bonding surfaces of specimens (ca.3 μm): (a) 33 W, F; (b) 37 W, F; (c) 43 W, F; (d) 40 W, L; (e) 46 W, L; (f) 50 W, L.

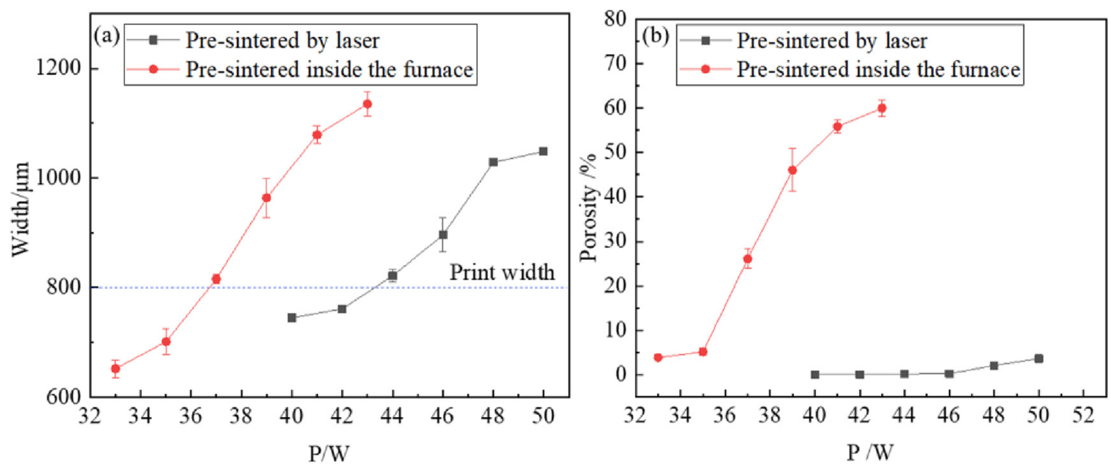

Fig. 14 – The effects on bonding width and porosity of different laser power (ca.3 μm): (a) bonding width; (b) porosity.

sintered by laser achieved the same bonding width required higher laser power than the samples pre-sintered inside a furnace. This is mainly because the frit layer pre-sintered by laser has a more compact structure with fewer pores and thus needs more laser energy to be melted fully.

The effect on the porosity of different laser power is shown in Fig. 14(b), the porosity of laser pre-sintered specimens increased slightly with the the increase of laser power. The porosity was only 3.59% when the laser power was 50 W. The porosity of the specimens pre-sintered in a furnace increased rapidly when the power was more than 35 W. The increase rate of porosity decreased at 41 W, but the porosity had reached 55.80%. The reason for the large porosity on specimen pre-sintered in a furnace can be explained by the changing bonding front, as shown in Fig. 15(a)–(c). The bonding frontier was curved and the melt area (1) had a crescent shape. In the melt area (2), the micro pores formed. The pores can be formed by the gases in pin-holes, or by the decomposition of the components of the glass frit. And the melt area (2) is located in the central area of the laser, the thermal expansion coefficient of the gas is much larger than that of the molten glass frit, and the micro pores would grow rapidly under the intense heat. But for the specimens pre-sintered by laser directly or re-sintered by laser, as shown in Fig. 15(d)–(f), both

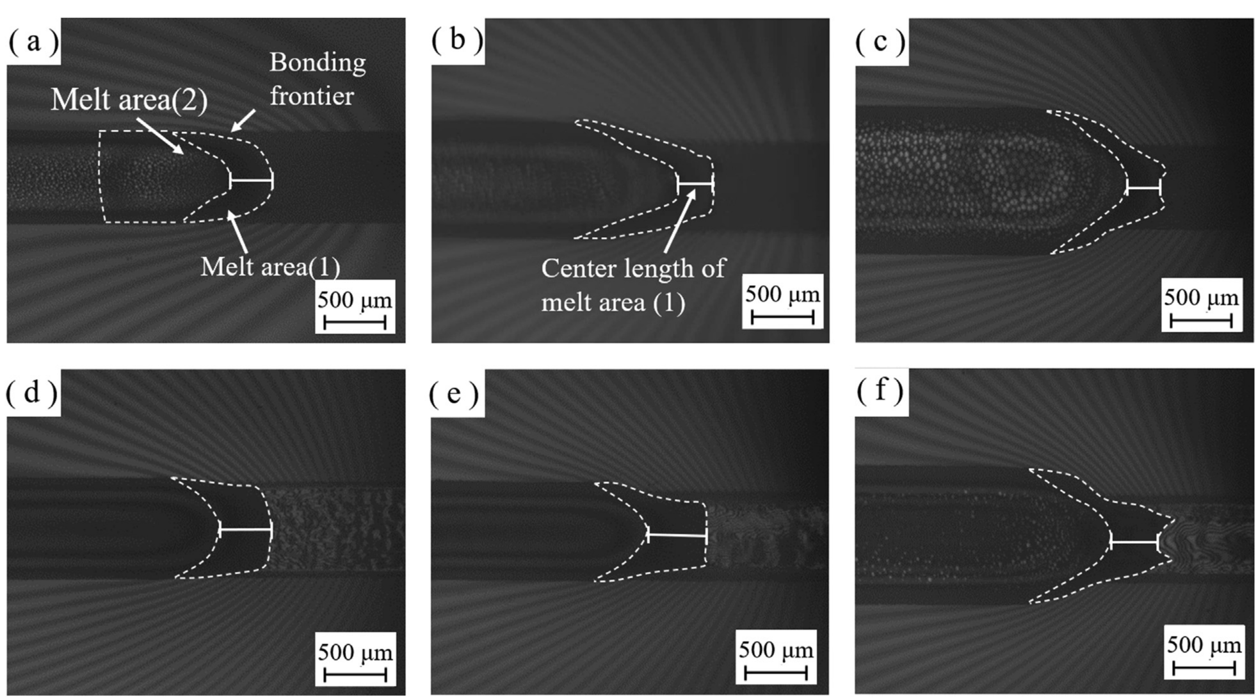

Fig. 15 – Laser bonding process photographed by CCD: (a) ca.3 μm, 37 W, F; (b) ca.8 μm, 35 W, F; (c) ca.8 μm, 41 W, F; (d) ca.3 μm, 44 W, L; (e) ca.8 μm, 36 W, F-L; (f) ca.8 μm, 40 W, F-L.

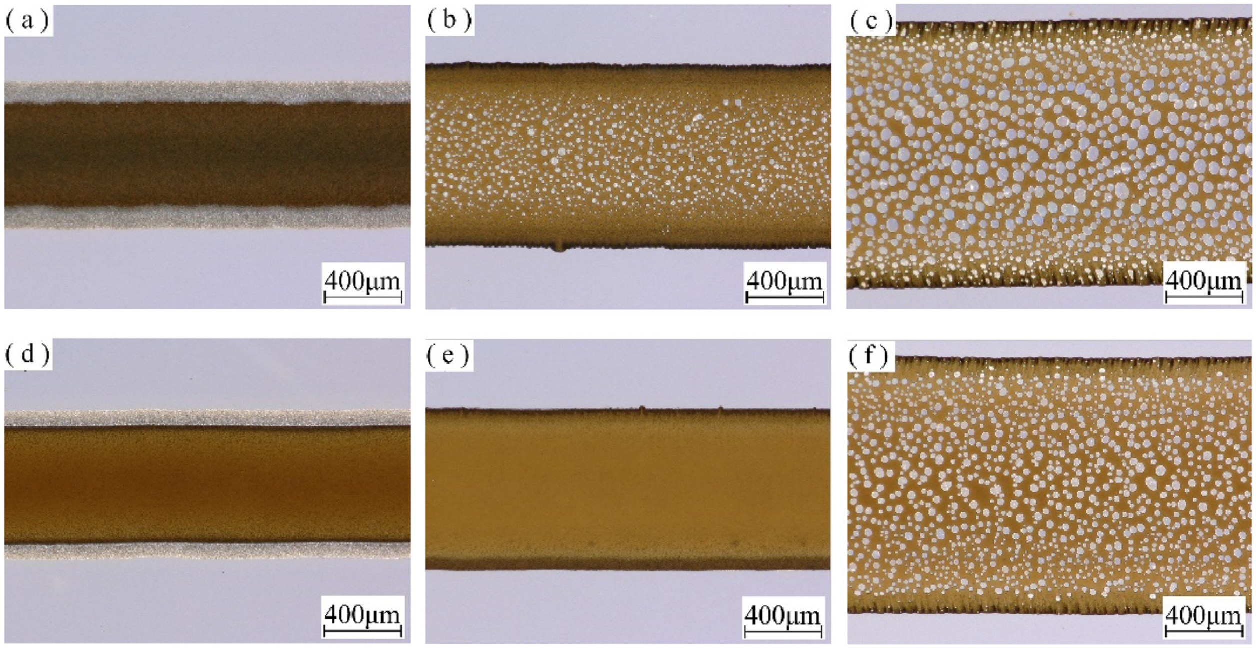

Fig. 16 – Surface of the bond in each specimen (ca.8 μm): (a)29 W, F; (b)35 W, F; (c)41 W, F; (d) 32 W, F-L; (e) 36 W, F-L; (f) 42 W, F-L.

sides of the bonding frontier were closer to the edge of the glass frit layer and the melt area (1) was larger, the bubbles were easier to overflow. Furthermore, they have denser frit layers and less decomposition than those of the specimens pre-sintered only by the furnace, and so it is difficult to form pores in the melt area (2). It is concluded that the laser pre-sintering or re-sintering could dramatically inhibit the porosity in the bonding process.

The bonding surface morphology of specimens (ca.8 μm) in parameter group 3 and parameter group 4 are shown in Fig. 16. For the furnace specimen, when the laser power was 29 W, only the middle part was connected, and the bonding width was 530 μm. There were no pores to appear. The gradient on both sides of the glass frit layer before bonding was small, and the width of the contact surface with the glass substrate is narrower, resulting in the bonding width being narrower than that in Fig. 13(a). When the power increased to 35 W, the glass frit layer was completely connected, and a large number of pores appear in the middle of the bonding area, with a porosity of 12.45%. When the laser power increased to 41 W, the width of the connection area increased to 1443 μm. Because the thickness of the glass frit layer increased, it is easier to expand after melting, so the maximum bonding width was wider. Pores were distributed densely in the bonding area, and the porosity was 40.63%. There is not a smooth saddle surface in the frit layer before bonding, and the contact was more sufficient, so pores were less than that of ca.3 μm.

For the specimens by laser re-sintering (26 W), as shown in Fig. 16(d)–(f), when the laser bonding power was 32 W, only the middle part was connected, and the connection width was

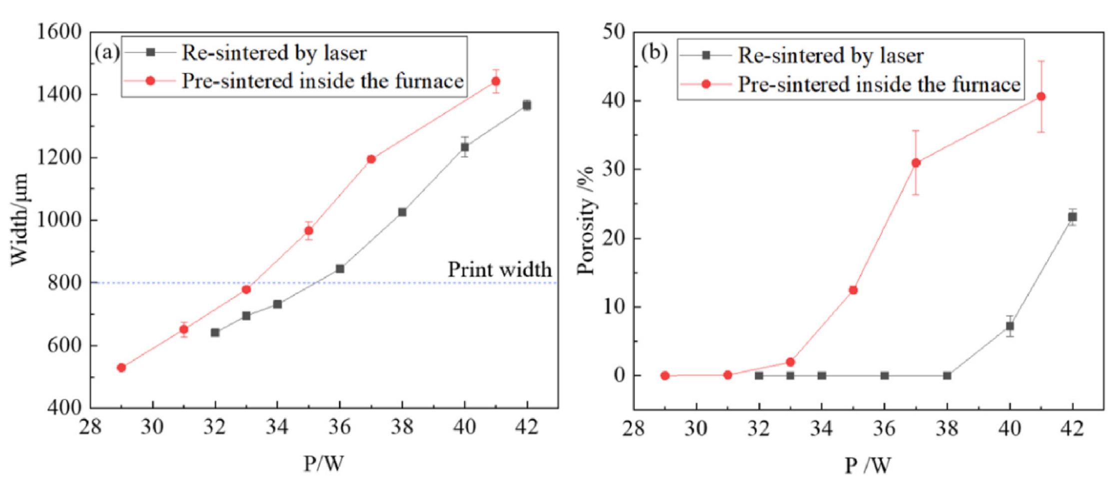

Fig. 17 – The effects on bonding width and porosity of different laser power (ca.8 μm): (a) bonding width; (b) porosity.

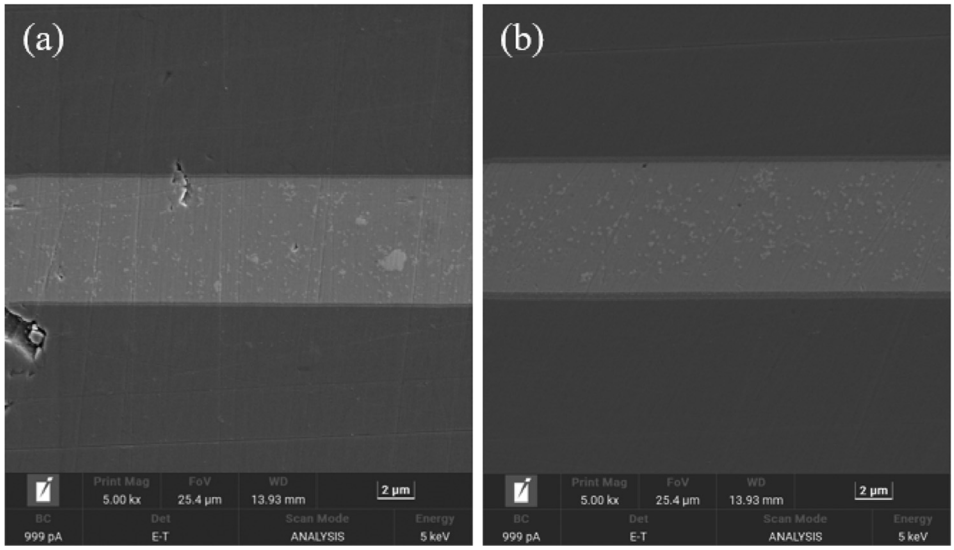

Fig. 18 – The results of SEM of the cross-section (ca.8 μm): (a) 35 W, F; (b) 36 W, F-L.

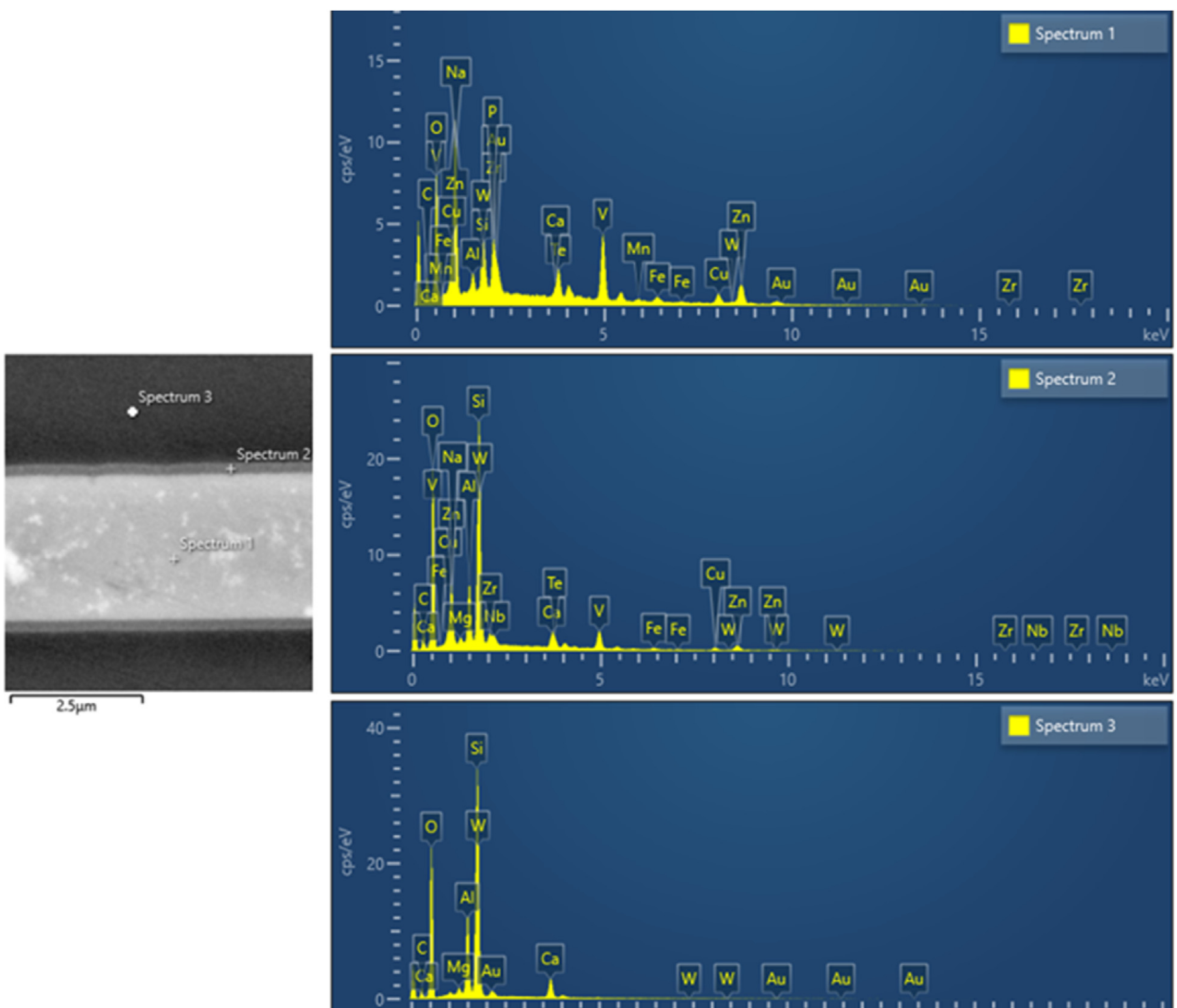

Fig. 19 – The results of EDS analyses of the cross-section (ca.8 μm): 36 W, F-L.

Table 3 – EDS point-scan analyses of the cross-section.

Element	Spectrum 1 Wt.%	Spectrum 2 Wt.%	Spectrum 3 Wt.%
Al	1.15	4.14	7.51
Si	1.58	15.03	22.90
Ca	0.55	1.78	4.14
V	11.82	4.73	0
Zn	21.31	8.85	0
Cu	7.75	3.25	0

641 μm, the connection area was relatively narrow compared with the sample in Fig. 13(d). And the pores were not visible. When the power increased to 36 W, the glass frit layer was completely connected and the melt area was diffused to both sides. There were no pores to appear. When the laser power increased to 42 W, a large number of pores appeared and densely distributed in the bonding area, with a porosity of 23.08%.

The center length of the melt area (1) was reduced with the increase of laser bonding power, as shown in Fig. 15(b) and (c), from 298 μm to 248 μm, more micro pores formed. The micro pores grew larger in melt area (2) with higher laser power, and the porosity was further increased. So, the porosity of the specimens pre-sintered in the furnace increased rapidly after 35 W, and reached 40.63% at 41 W, as shown in Fig. 17. For specimens by re-sintering, the center length of the melt area (1) (Fig. 15(e) and (f), from 437 μm to 374 μm) was longer. Before the micro pores formed, the molten glass frit had a longer distance to fill the gap between the glass substrates. The porosity increases rapidly when laser power increased to 40 W, and reached 7.22%. The width of the front of the melt area (1) was only 582 μm, but the bonding width increased to 1234 μm rapidly. So, more pores appeared with the rapid expansion of molten glass frit. When the laser power was lower than 38 W, the formation of pores can be effectively restrained by laser re-sintering.

3.4. EDS image analysis

The SEM results of the cross-section (ca.8 μm) are shown in Fig. 18. During the bonding process, the glass substrates and glass frit penetrated each other, and a banded area appears at the fusion interface, which means that a new glass phase appeared. When the laser power was 35 W, the sample pre-sintered in a furnace had a glass frit layer thickness of 6.67 μm and a new glass phase thickness of 0.17 μm. The sample re-sintered by laser had a thicker glass frit layer (6.92 μm, power 36 W). Because of the longer center length of the melt area (1), as shown in Fig. 15, it had a thicker new glass phase (0.37 μm), and more penetration on the surface of glass substrates is observed.

In Fig. 19, EDS was used to detect the bonding section. The substrate, glass frit and fusion interface were scanned point-by-point, respectively, and the result was as shown in Table 3. The unique components V, Zn and Cu in the glass frit penetrated the glass substrate, while Si, Al and Ca in the glass substrate penetrated the glass frit.

3.5. Bonding energy

Bonding energy can be used to quantify the strength of the created connections. The bonding energy is measured by separating the bonded glass plates by cleaving, and calculated per unit area just like the material constant of surface energy. To estimate the bonding energy, a test setup based on Gillis and Gilman's theoretical analysis of crack propagation [21,22] had been assembled with the test schematic, as shown in Fig. 20.

By inserting a blade of thickness $d_b = 50$ μm into the edge of the welded sample and measuring the length of the gap (L) till the joint is destroyed, the bonding energy can be calculated from the following equation:

$$E_b = \frac{3Ed_b^3 d_p^2}{32L^4} \quad (2)$$

where E is the elastic modulus of the glass material ($E = 73.6$ GPa) and $d_p = 0.7$ mm is the thickness of a glass plate. It should be pointed out that E_b represents the bonding energy of the weld seam itself without the optical contact contribution. The bonding energy is the energy when the glass substrates are just cracked by the blade.

For the specimens (ca.8 μm) re-sintered by laser, with the increase of laser power, the bonding energy increased until

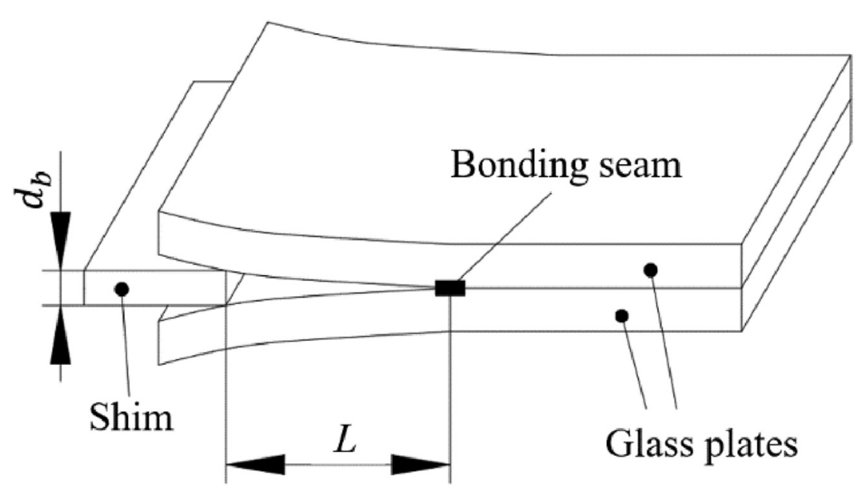

Fig. 20 – Schematics of the gap opening experiment to determine the bonding energy.

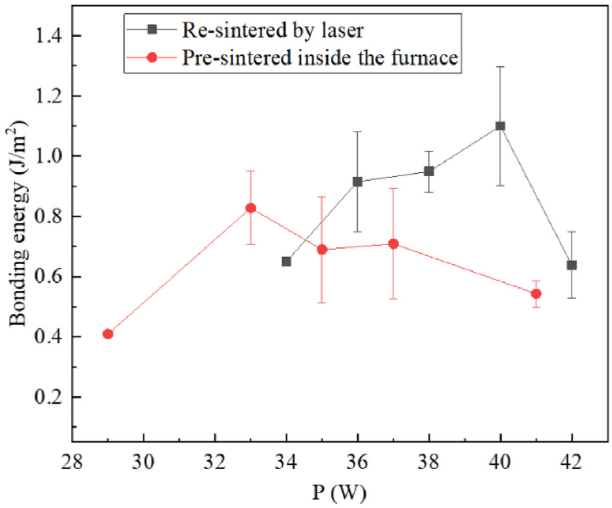

Fig. 21 – The effect on Bonding Energy of different laser power (ca.8 μm).

the laser power was 40 W, and the bonding energy reached the peak value of 1.1 J/m^2, as shown in Fig. 21. Because the increase of the bonding width could lead to an increase in the bonding energy, a small number of pores have little effect on the bonding strength. However, when the porosity exceeded 20% (power 42 W), the bonding energy was reduced to 0.64 J/m^2. For the specimens pre-sintered inside the furnace, when the laser power was 33 W, the bonding width (779 μm) approached the glass frit width, and the bonding energy reached the peak value of 0.83 J/m^2. Because of the rapid increase of the porosity, the bonding energy didn't increase with the increase of the bonding width. When the porosity was 40.2%, the bonding energy was reduced to 0.54 J/m^2. It is found that the treatment of re-sintering by laser can improve the bonding strength due to more penetration and less porosity.

4. Conclusions

In this work, the bonding performance of the specimens prepared by three pre-sintering methods was studied. The laser pre-sintering and bonding effect were analyzed in detail.

(1) Although the laser pre-sintering process is more efficiency than furnace sintering, the surface of pre-sintering by laser is easy to form small pits. Fortunately, the later laser bonding could remove such small pits.
(2) For the specimens pre-sintered by laser with low thickness of glass frit layer (ca.3 μm), the porosity is obviously inhibited. Both sides of the bonding frontier were closer to the edge of glass frit layer and the laser pre-sintering can produce a compact pre-sintering layer.
(3) The specimens with large frit thickness (ca.8 μm) need to be pre-sintered in a furnace first and then re-sintered by laser, in order to inhibit the pores which formed in the process of laser bonding effectively.
(4) The laser re-sintered process can lead to longer center length of the melt area (l) and more penetration. Because of more penetration and less porosity, the treatment of re-sintering by laser can improve the bonding strength significantly.

Author Contribution

Peixin Zhong: Writing–original draft, Investigation, Formal analysis. **Genyu Chen**: Supervision. **Shaoxiang Cheng**: Investigation, Validation. **Mingquan Li**: Writing–review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful for the financial support provided by the National Key R&D Program of China (project number: 2017YFB1104802).

REFERENCES

[1] Knechtel R. Glass frit bonding: an universal technology for wafer level encapsulation and packaging. Microsyst Technol 2005;12:63–8. https://doi.org/10.1007/s00542-005-0022-x.
[2] Granados L, Morena R, Takamure N, Suga T, Huang S, McKenzie DR, et al. Silicate glass-to-glass hermetic bonding for encapsulation of next-generation optoelectronics: a review. Mater Today 2021;47:131–55. https://doi.org/10.1016/j.mattod.2021.01.025.
[3] Tian R, Li Y, Yin L, Zhang J. Hermeticity test of low-melting point sealing glass and analysis of encapsulation failure. 18th Int. Conf. Electron. Packag. Technol. 2017;2017:844–9. https://doi.org/10.1109/ICEPT.2017.8046577.
[4] Ramm P, Lu JJQ, Taklo MMV. Handbook of wafer bonding, 2012. Wiley VCH; 2012. https://scholar.google.com/scholar_lookup?title=Handbook%20of%20Wafer%20Bonding&publication_year=2012&author=P.%20Ramm&author=j%20j%20q%20Lu&author=M.M.V.%20Taklo.
[5] Kim DN, Lee J, Huh JS, Kim H. Thermal and electrical properties of BaO–B_2O_3–ZnO glasses. J Non-Cryst Solids 2002;306(1):70–5. https://doi.org/10.1016/S0022-3093(02)01050-5.
[6] Dresbach C, Krombholz A, Ebert M, Bagdahn J. Mechanical properties of glass frit bonded micro packages. Microsyst Technol 2006;12(5):473–80. https://doi.org/10.1007/s00542-005-0031-9.
[7] Emami S, Martins J, Andrade L, Mendes J, Mendes A. Low temperature hermetic laser-assisted glass frit encapsulation of soda-lime glass substrates. Opt Laser Eng 2017;96:107–16. https://doi.org/10.1016/j.optlaseng.2017.04.006.
[8] Emami S, Martins J. R Madureira, D Hernandez, G Bernardo, J Mendes, A Mendes, Development of hermetic glass frit encapsulation for perovskite solar cells. J Phys D Appl Phys 2018;52(7):074005. https://doi.org/10.1088/1361-6463/aaf1f4.

[9] Yadav A, Joshi J, Singh DK, Natu H, Rotti C, Bandyopadhyay M, et al. Development of laser beam welding for the lip seal configuration. Fusion Eng Des 2015;96–97:192–8. https://doi.org/10.1016/j.fusengdes.2015.05.064.

[10] de Pablos-Martin A, Benndorf G, Tismer S, Mittag M, Cismak A, Lorenz M, et al. Laser-welded fused silica substrates using a luminescent fresnoite-based sealant. Opt Laser Technol 2016;80:176–85. https://doi.org/10.1016/j.optlastec.2016.01.020.

[11] Kim SY, Park J, Kim SH, Kadathala L, Baek JH, Kim JH, et al. The tuning capability of CuO and Na_2CO_3 dopant on physical properties for laser sealing using fiber. Appl Sci 2020;10:353. https://doi.org/10.3390/app10010353.

[12] DK Ivanou R Santos, Maçaira J, Andrade L, Mendes A. Laser assisted glass frit sealing for production large area DSCs panels. Sol Energy 2016;135:674–81. https://doi.org/10.1016/j.solener.2016.06.043.

[13] Xiao Y, Wang W, Wu X, Zhang J. Process design based on temperature field control for reducing the thermal residual stress in glass/glass laser bonding. Opt Laser Technol 2017;91:85–91. https://doi.org/10.1016/j.optlastec.2016.12.016.

[14] Emami S, Martins J, Ivanou D. Advanced hermetic encapsulation of perovskite solar cells: the route to commercialization. J Mater Chem 2020;8(5):2654–62. https://doi.org/10.1039/c9ta11907h.

[15] Tian R, Yin L, Li Y, Zhang J. The effect of glass frit paste levelling property on encapsulation. In: 19th International conference on electronic packaging technology (ICEPT). 2018. IEEE; 2018. p. 1097–101. https://doi.org/10.1109/ICEPT.2018.8480596.

[16] Lorenz N, Millar S, Desmulliez M. Hermetic glass frit packaging in air and vacuum with localized laser joining. J Micromech Microeng 2011;21(4):045039. https://doi.org/10.1088/0960-1317/21/4/045039.

[17] Beerkens RGC, van der Schaaf J. Gas release and foam formation during melting and fining of glass. J Am Ceram Soc 2006;89(1):24–35. https://doi.org/10.1111/j.1551-2916.2005.00691.x.

[18] Zhang L, Sharif A, Yeung H, Rong E, Riko IM, Long L, et al. Ceramic-Ceramic joining using glass frit for high temperature application. 2012 IEEE 14th Electronics Packaging Technology Conference (EPTC). IEEE; 2012. p. 38–42. https://doi.org/10.1109/EPTC.2012.6507047.

[19] Liu Y, Chen D, Lin L, Zheng G, Zheng J, Wang L, et al. Glass frit bonding with controlled width and height using a two-step wet silicon etching procedure. J Micromech Microeng 2016;26:035018. https://doi.org/10.1088/0960-1317/26/3/035018 (8pp).

[20] Kind H, Gehlen E, Aden M, Olowinsky A, Gillner A. Laser glass frit sealing for encapsulation of vacuum insulation glasses. Phys Procedia 2014;56:673–80. https://doi.org/10.1016/j.phpro.2014.08.075.

[21] Kalogirou S. Sol energy eng: process syst. ; 2009. https://books.google.com/books?hl=ja&lr=lang_ja|lang_en&id=wYRqAAAAQBAJ&oi=fnd&pg=PP1&dq=Kalogirou+S.+Sol+Energy+Eng:+Process+Syst+2009.&ots=LbEVAXJOJS&sig=ywR_z5EVFniis-A7QiHMPZc2oFA.

[22] Alexeev I, Cvecek K, Schmidt C. Characterization of shear strength and bonding energy of laser produced welding seams in glass. J. Laser Micro Nanoeng. 2012;7(3). https://doi.org/10.2961/jlmn.2012.03.0009.